Applied Catalysis A: General 526 (2016) 45–52

Contents lists available at ScienceDirect

Applied Catalysis A: General

journal homepage: www.elsevier.com/locate/apcata

Effect of confinement of TiO_2 nanotubes over the Ru nanoparticles on Fischer-Tropsch synthesis

Xu Yang [a], Wenbo Wang [a], Liangpeng Wu [a], Xinjun Li [a,*], Tiejun Wang [a,*], Shijun Liao [b]

[a] Key Laboratory of Renewable Energy, Guangzhou Institute of Energy Conversion, Chinese Academy of Sciences, Guangzhou 510640, China
[b] School of Chemistry and Chemical Engineering, South China University of Technology, Guangzhou 510641, China

ARTICLE INFO

Article history:
Received 28 March 2016
Received in revised form 19 July 2016
Accepted 22 July 2016
Available online 3 August 2016

Keywords:
Confinement
TiO_2 nanotubes
Ruthenium
Fischer-Tropsch synthesis

ABSTRACT

Ru nanoparticles (NPs) have been successfully deposited in the inner pores of TiO_2 nanotubes (TNTs), denoted as Ru-in/TNT, by vacuum assisted impregnation method for Fischer-Tropsch synthesis (FTS). The composition, morphology and electronic properties have been characterized by the XRD, TEM, N_2-sorption, H_2-TPR/TPD and XPS, investigating the confinement effect of TNT on the entrapped Ru NPs' FTS performance. The Ru-in/TNT exhibits high CO conversion activity and superior deactivation-resist ability over the Ru-out/TNT (with most of Ru NPs deposited on the outer surface of TNT) and Ru/P25 (using commercial P25 powder as support) catalysts. Besides above superiority, the Ru-in/TNT facilitates the formation of long chain (C19+) products as well. The enhanced activity and selectivity can be ascribed to the confinement effect of TNT: (i) strengthen charge transfer and modulate the electronic properties of Ru species resulting in more active center site; (ii) constrain the metal sintering and metal leaching to resist deactivation during FTS; (iii) facilitate the re-adsorption of α-olefin leading to long-chain product.

1. Introduction

Fischer–Tropsch synthesis (FTS) is a key step for the transformation of nonpetroleum feedstocks (natural gas, coal, and biomass) into hydrocarbon fuels by syngas (CO/H_2). Various supported catalysts have been widely investigated over the last 80 years or more [1–3].

It is well established that Ru is the most active metal for CO hydrogenation with higher CO/H_2 conversion rates and higher average molecular weight hydrocarbon yields as compared to Fe- or Co-based catalysts [4,5]. Moreover, ruthenium can work under higher partial pressures of water or other oxygenate-containing atmospheres, which makes it practically important for the conversion of syngas produced from biomass. However, the high initial activity displayed by the Ru supported catalysts during FTS often decreases over time on stream due to several causes. For example, oxidation of the metallic particles, formation of volatile metallic oxides, metal agglomeration, leaching of the active phase, carboneous coking, etc would somehow cause catalytic activity decline to various extent. To address above issues, researchers have developed various method to improve the performance of Ru catalysts. One effective protocol is adding promoter to the support of Ru component: Nurunnabi and et al. [6] have found that addition of Mn to gamma-Al_2O_3 could enhance the removal of chloride from $RuCl_3$, which can lead to the formation of moderate sized Ru particles with improved FTS activity. By adding second metal (Co, Fe and etc.) to form M-Ru alloy [7]: Li and co-workers [8] have reported that small portion of Fe species alloyed with Ru can stablize the Fe-Ru bimetallic NPs. Another efficient strategy is to constrain the active Ru species within confined space, such as porous, lamller, core-shell and other unique structures. Sadakiyo and co-workers [9] have prepared porous metal organic framwork ZIF-8 supported Ru catalyst. In Li's group [10], they have found that the Ru emmbed in carbon nanotube could supress particle aggregation, movement and oxidation, exhibiting higher stablity than the activated carbon supported Ru catalyst.

Morever, entrapping the active components in a confined space has been proved to be an efficient method to improve the catalyst's performance since the pioneer work by Bao's group [11]. They have declared that the confinement effect of carbon nanotube on the entrapped active component can hinder the metal sintering process, strengthen the electron transfer and significantly improve the activity and stablity of the catalysts [12–14]. Based on the TiO_2 nanotubes (TNT)'s two promising strucutral attributes (tubular inner pores and metal oxide properties), we have also reported a series of novel highly-dispersed Pt, Pd and Au nano-particles (NPs)

* Corresponding authors.
E-mail addresses: lixj@ms.giec.ac.cn (X. Li), wangtj@ms.giec.ac.cn (T. Wang).

http://dx.doi.org/10.1016/j.apcata.2016.07.021

entrapped in TNT catalysts, and demonstrated that confining the metal NPs within TNT could be an effective way to improve the catalyst's performance as well [15,16].

Herein, as continuation of our TNT confined metal NPs catalyst work, we attempted to confine the ruthenium NPs within the TNTs via vacuum-assisted impregnation route to obtain a high performance Ru catalyst, Ru-in/TNT, for FTS. The confinement effect of TNT on the FTS performance of the entrapped Ru NPs was comprehensively investigated.

2. Experiment

2.1. Catalysts preparation

TiO_2 nanotubes (TNTs) were synthesized by a previously described method [17]. As-received TiO_2 (Hualisen Corp., China, CP) was mixed with 100 mL concentrated NaOH (Sinopharm. Corp., China, AR) solution (10 M) in a Teflon container under mild stirring at 140 °C for 24 h; the resultant white precipitate was then separated from the mixture and rinsed with deionized water several times until the conductivity of the solution drops below 100 μs/cm and keeps constant. The solution conductivity was measured by a DDS-307 Electric Conductometer (Leici, China).

TNT-supported ruthenium catalysts were prepared according to our previous reported procedures [16]. First, the TNT support was impregnated via the vacuum assisted incipient wetness impregnation technique, using an aqueous solution containing a calculated amount of $RuCl_3$ (Sinopharm co., China, AR) under pressure of <10^{-2} pa; the mixture was then vacuum dried overnight at 40 °C, followed by calcination at 300 °C for 2 h. This novel catalyst was denoted as Ru-in/TNT, with fixed Ru loading of ca. 5.0 wt%. A referenced Ru catalyst, with the same metal loading, was prepared by the typical wet-impregnation method using TNT and commercial P25 (Degussa, Hualisen Corp., China) powder as support, and denoted as Ru-out/TNT and Ru/P25 respectively, seeing Table 1.

2.2. Catalyst characterization

X-ray diffraction (XRD) patterns were obtained with an X'Pert Pro MPD X-ray diffractometer (PANalytical, Netherlands) using Cu Kα radiation. N_2 adsorption-desorption isotherms were measured with a Tristar 3010 isothermal nitrogen sorption analyzer (Micromeritics, USA) using a continuous adsorption procedure. Transmission electron microscopy (TEM) was carried out on a JEM-2010 microscope (JEOL, Japan) using an accelerating voltage of 200 kV. X-ray photoelectron spectroscopy (XPS) was performed with an AXIS Ultra DLD (Kratos, Britain) to examine the catalysts' electronic properties. Hydrogen temperature-programmed reduction/desorption (H_2-TPR/TPD) was performed on an Autosorb-iQ-C automated system (Quantachrome, USA). Approximate 50 mg of as-prepared catalyst was placed in a quartz tube within a temperature-controlled oven. H_2-TPR was conducted at a heating rate of ca. 10 °C min^{-1} from 35 to 800 °C. For H_2-TPD, the catalyst was first reduced in H_2 flow at 100 cm^3 min^{-1} for 1 h at 400 °C and then cooled to room temperature. TPD was performed using a constant heating rate of ca. 10 °C min^{-1} from 35 to 600 °C. Inductively coupled plasma atomic emission spectrometry (ICP-AES) was conducted by a Perkin Elmer Optima 3300 DV ICP-AES (Perkin Elmer, Canada) to determine the catalysts' Ru content.

2.3. Fischer-Tropsch synthesis

The catalytic properties of the as-prepared samples were performed in a fixed bed flow reactor. According to the procedure reported elsewhere [18], 1.0 g catalyst was firstly reduced with H_2 at 300 °C for 12 h. After that, changing the input gas from H_2 into syngas (H_2/CO = 1.0, space velocity of 3000 h^{-1}), the pressure was elevated to 2.0 MPa, and the reaction temperature was maintained at given temperate (200–260 °C). The production selectivity changed slightly in this temperature range (seeing Table S1 in Supporting information), thus all of the FTS reactions in this text were performed at medium temperature of 240 °C if not mentioned especially. The mass balance was established for all performance runs. Agilent 7890A was used to detect H_2, CO in the feed gas and H_2, CO, CO_2 and hydrocarbon (C_1–C_4) in tail gas. The liquid products were collected and weighed, and then detected using gas chromatographs (GC2010, Shimadzua). The error in mass balance was likely to be resulted from the inaccuracy during the measurements of the streams, but would not exceed ±10%. In addition, the hydrocarbon distribution was expressed as the weight percentage of desired components in all the hydrocarbons.

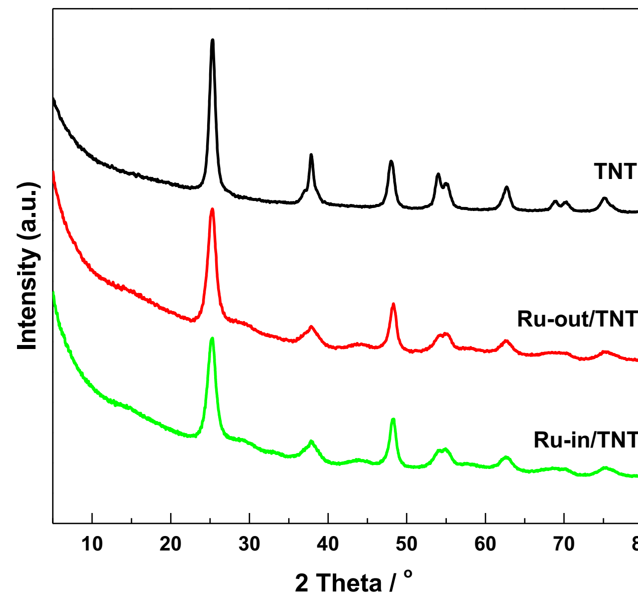

Fig. 1. XRD pattern of various catalysts.

3. Results

Table 1 gives the porosity parameters for the Ru catalysts used in this text. The TNT based Ru catalysts have high surface area (~230–240 m^2/g) with pore size of 7–8 nm, and decreased slightly after 48 h FTS reaction. It implies the good thermal stability of TNT. High surface area would facilitate the metal dispersion and mass transfer.

Fig. 1 shows the XRD patterns of the samples reduced at 300 °C. A group of peaks centered at 25, 37, 48, 55 and 62° of 2θ are observed for the TNT support, corresponding to the typical anatase crystallite of the support after thermal treatment. For the Ru loaded samples, no peaks account for the metallic Ru compounds (PDF #06-0663) are detected in the XRD patterns [19,20]. It suggests that these species exist either in an amorphous state or with particle sizes smaller than the XRD limit (<4 nm); there is no apparent difference of the XRD data between the Ru-out/TNT and Ru-in/TNT.

Fig. 2 shows the TEM images of the fresh catalysts. Both the Ru-out/TNT and Ru-in/TNT present a high dispersion of Ru with particle size range of 2–4 nm. For the Ru-in/TNT, the entrapped Ru NPs in the inner pore of TNT are visible in Fig. 2c and e. While most part of the Ru NPs are deposited on the outer surface of TNT for the Ru-out/TNT, seeing Fig. 2d and f. The fringe value presented in Fig. 2f is calculated to be 0.20 nm, which agrees well with the $d_{(101)}$ plane spacing of metallic Ru. This result confirms that the Ru NPs can

Table 1
Specifications of catalysts.

CATA.	Ru wt.%[a]	S_{BET} (m^2/g)[b]	D_{Pore} (nm)[b]	D_{Ru} (nm)[c]	V_{H2} (mmol/g)[d]
Ru-out/TNT	5.0	232 (200)*	8.1	3.2 (5.0)*	0.381
Ru-in/TNT	4.8	240 (205)	7.9	2.8 (4.0)	0.464
Ru/P25	5.1	20 (18)	–	4.2	0.234

*The corresponding surface area and particle size crystallite sizes of the spend catalysts are given in parentheses for comparison.
[a] Ru content was determined by the ICP.
[b] Surface area (S_{BET}) and pore diameter (D_{Pore}) were derived by nitrogen isothermal sorption analysis.
[c] Average values were calculated from 200 individual crystallites in TEM images.
[d] H_2 adsorption was determined from the area integration of TCD signals in H_2-TPD.

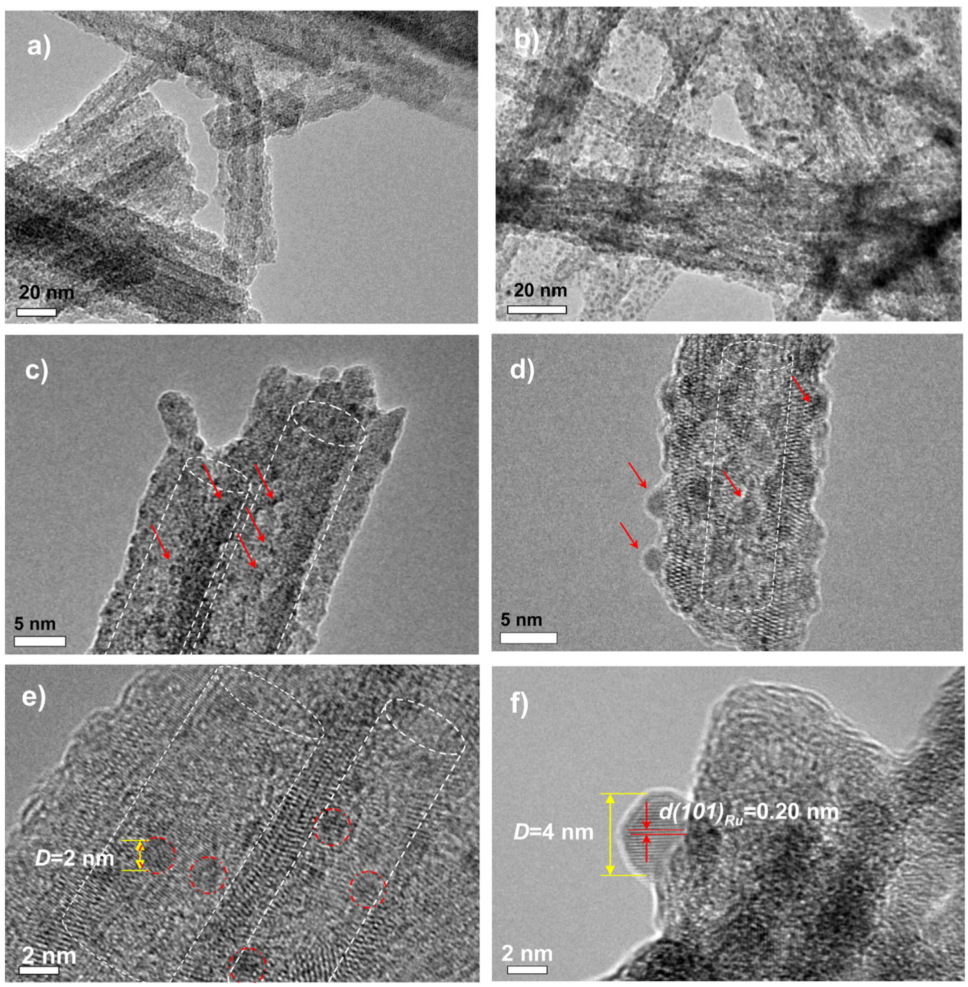

Fig. 2. High resolution TEM images of Ru-in/TNT (a, c, e) and Ru-out/TNT (b, d, f).

be confined in TNT via vacuum-assisted impregnation method. The corresponding Ru average particle size values are in the sequence of Ru-in ≈ Ru-out > Ru/P25, seeing Table 1, indicating that the metal dispersion is associated well with the surface area of support.

Fig. 3 shows the TEM images of the spend catalysts after 48 h FTS reaction. The tubular morphology of TNT was preserved well (Fig. 3a and b). It explains well the reason for the little decrease of surface area for the used TNT supported catalyst. Compared with the fresh ones, the Ru NPs of the spend catalysts have somehow aggregated severely for the Ru-out/TNT catalysts (Fig. 3c and e), indicating that the Ru-in/TNT possesses higher sintering-resist ability than the Ru-out/TNT (Fig. 3d and f).

XPS measurement was conducted to investigate the surface properties and chemical states of the catalysts. The peaks for C 1s and C-OR are centered at 284.8 and 288.4 eV (Fig. 4a and b), which is ascribed to the surface carbonaceous contamination. A relatively low peak at ~280.1 eV assigned to the Ru 3d spectrum can be fitting into various curves indicating various Ru species. It can be seen that the Ru exists in both Ru^{III} and Ru^0 phases in Ru-out/TNT while a single metallic phase in Ru-in/TNT. This implies the various redox properties of Ru deposited on inner or outer pores of TNT. Moreover, compared with the Ru-out/TNT sample, a peak shift of 0.3 eV towards high energy can be observed for Ru-in/TNT. Correspondently, a negative shift (0.1 eV) of Ti 2p occurred for Ru-in/TNT as well (Table 2). These results indicate the strong interaction between Ru particles and TiO_2 framework, which could promote the efficient electron transfer from the conduction band of TiO_2 to the Ru particles similar to the case of Au/TiO_2 [21]. Actually it is an evident that confinement of TNT can modify the electronic properties of the entrapped Ru. The integral peak of O1s decomposes into two components, oxygen species with binding energy of 530.0 and 532. eV, corresponds to the oxygen in the composition of the three dimen-

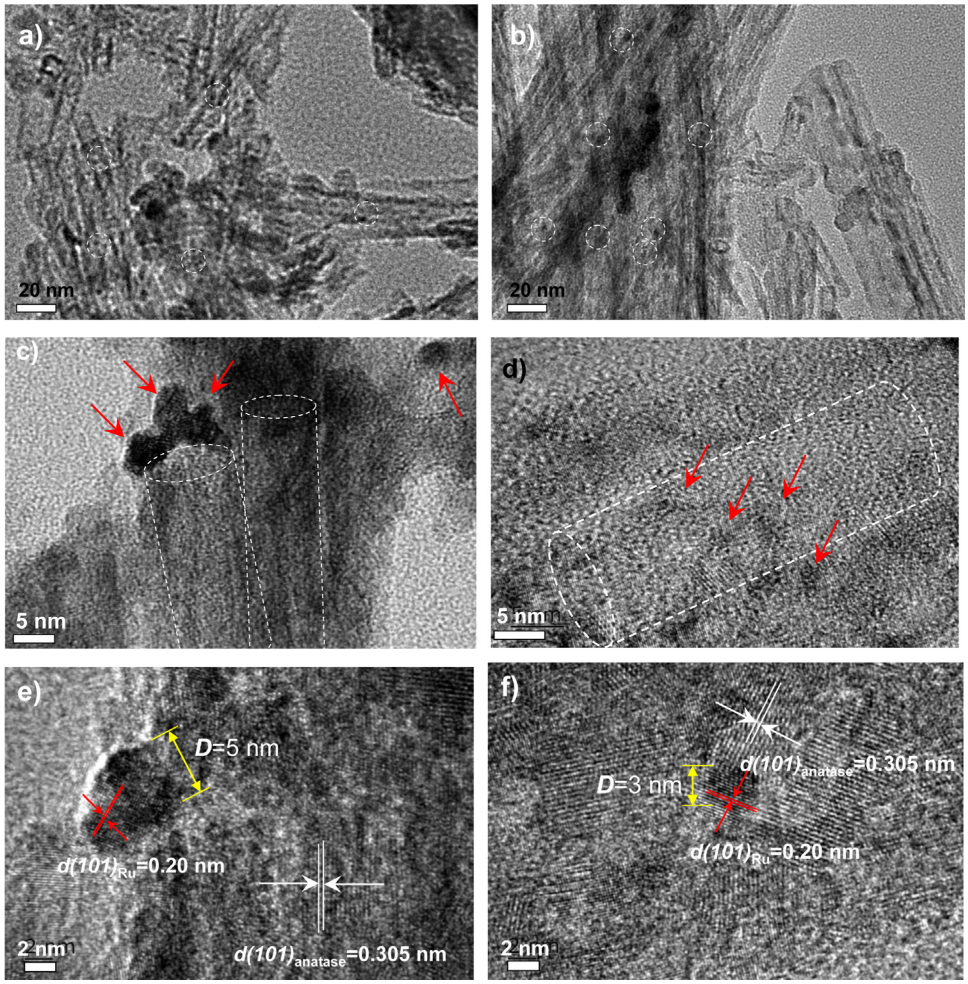

Fig. 3. TEM images of the spend catalyst: Ru-out/TNT (a, c, e) and Ru-in/TNT (b, d, f).

Table 2
XPS analysis of fresh catalysts: binding energy and atomic ratio.

Catalyst	Binding energy (eV)					Atomic ratio
	$Ru^0{}_{3d5}$	$Ru^{II}{}_{3d5}$	Ti_{2p3}	C_{1s}	O_{1s}	Ru/Ti
Ru-out-TNT	280.1	–	458.7	289.8	530.4	0.24
Ru-in-TNT	280.4	281.3	458.6	289.8	530.4	0.25

Table 3
XPS analysis of spend catalysts: binding energy and atomic ratio.

Catalyst	Binding energy (eV)					Atomic ratio
	$Ru^0{}_{3d5}$	$Ru^{II}{}_{3d3}$	Ti_{2p3}	C_{1s}	O_{1s}	Ru/Ti
Ru-out/TNT	280.1	–	458.8	284.8	530.4	0.05
Ru-in/TNT	281.6	–	458.8	284.8	530.4	0.27

sional titanium oxide [22], and oxygen in the composition of the —OH group or water [23].

Fig. 5 shows the corresponding XPS spectra for the spend samples. It provides some important information of the two catalysts after 48 h of FTS: (i) the Ru phase in both of the two catalysts existed as metallic form, indicating that the oxidized Ru was fully reduced by the syngas; (ii) the shift of Ru binding energy in Ru-in/TNT becomes more prominent, up to 1.5 eV. The Ru-in/TNT holds more of O phase with binding energy of ~532 eV. Moreover, the atomic ratio of catalysts exhibits, seeing Table 3, the Ru-out/TNT declined remarkably, while the Ru-in/TNT preserved stable as the fresh one, demonstrating severely metal leaching occurred in Ru-out/TNT.

The H_2-TPR patterns of catalysts are presented in Fig. 6a. A sharp peak locates in the range of 150–200 °C for all catalysts indicating the reduction of Ru^{III} precursor ($RuCl_3$) or RuO_2 into metallic Ru. The reduction peak of Ru-in/TNT shifts slightly (ca. 7 °C) towards low temperature region relative to the Ru-out/TNT, indicating the discrepancy in reducibility for the two catalysts, which agrees well with the XPS analysis. In the corresponding H_2-TPD pattern of Fig. 6b, the Ru-out/TNT shows one main broadened peak in the range of 200–600 °C; the Ru-in/TNT exhibits two sharp peaks centered at 390 and 610 °C, respectively. Documents [24,25] have reported that the H_2-TPD peaks occurred at temperature over 100 °C can be ascribed to the strong adsorption of hydrogen on the highly dispersed Ru NPs. Moreover, during the desorption process at high temperature, these hydrogen atoms on the support would migrate back to the metal surface and then recombine to hydrogen molecules and cause spillover desorbing [26]. The Ru-in/TNT possesses higher H_2 desorption temperature, suggesting the stronger metal-support interaction on the entrapped Ru NPs. The amount of H_2 desorption, determined by integrating the area of these peaks in the TPD pattern (Fig. 6b), shows that the Ru-in-TNT possesses high hydrogen adsorption capacity (0.464 mmol/g), which is 20% and 100% higher than that of Ru-out/TNT and Ru/P25. This demonstrates that the Ru-in/TNT hold more of active center sites than the other two catalysts, which could influence the catalytic behavior of the CO on the catalyst consequently.

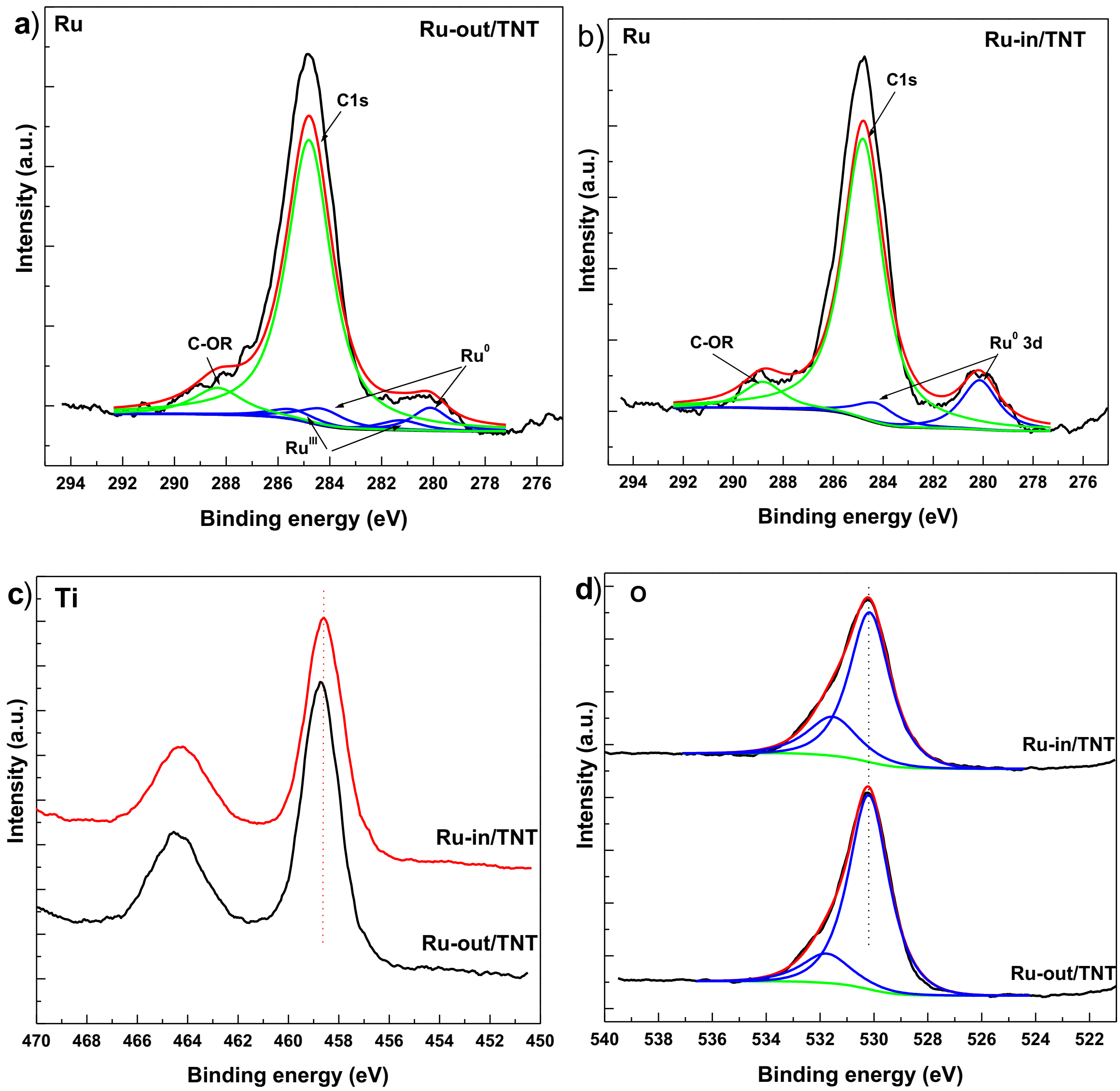

Fig. 4. XPS spectra of fresh catalysts.

The ruthenium catalysts reduced at 300 °C were tested in the Fischer-Tropsch reaction using syngas with a H_2/CO ratio of 1 (which is closed to the ratio produced from biomass [18]) to inspect the influence of the confinement of TNT on their activity and selectivity. As shown in Fig. 7, at the initial stage of the reaction, the Ru-in/TNT presented ~2 times higher CO conversion than that of Ru-out/TNT and Ru/P25, implying the high initial activity of the entrapped Ru NPs. As the reaction proceeded, all of the Ru based catalysts went through an obvious activity decline period (~12 h). This profile corresponds to the catalyst deactivation, which was also observed by other research groups over the Ru based catalyst in FTS [6,27]. After 20 h reaction, the activity of Ru-out/TNT and Ru/P25 showed nearly 95% decline with low CO conversion of 3% and 4%, respectively. However, for the Ru-in/TNT, after the first stage of decline the catalyst's activity kept stable and approached a relatively constant value of ca. 20% till the reaction was terminated. Indeed, a high CO conversion rate of 60 μmol/g_{Ru}s was achieved, which is higher than most part of Ru catalyst reported previously [28,29]. The activity enhancement of Ru-in/TNT can be ascribed to the improved metal dispersion and strengthened electronic effect between the entrapped Ru NPs and TNT. The TOFs values of the two catalysts (seeing Tables S2 and S3 in Supporting information) decreased after FTS reaction; the Ru-in/TNT exhibited higher TOFs than the Ru-out/TNT either at the initial (2 h) and stable (20 h) reaction stage, indicating that the confinement effect could enhance the entrapped Ru particles' activity. Noticeably, after FTS reaction, large amount of free Ru ions (0.35 mmol/g) were detected for the Ru-out/TNT, while few Ru ions (0.08 mmol/g) were observed in Ru-in/TNT (seeing Table S4 of Supporting information). This result accords well with the low Ru/Ti atomic ratio of used Ru-out/TNT in XPS analysis. It verifies that severely metal leaching has occurred in the Ru-out/TNT. This is can be the mainly reason for the large deactivation (~90%) of Ru-out/TNT after 48 h FTS reaction.

With respect to the product selectivity, seeing Fig. 8 and Table 4, all of the Ru catalysts have a similar CH_4 selectivity (~21–22%), while the Ru-in/TNT has an relative higher C_{5+} selectivity, especially for the C_{19+} hydrocarbon. Accordingly, the Ru-in/TNT has lower olefin/paraffin molar ratio than the other two catalysts. Previous literatures [30,31] have reported that the particle size, support properties, reaction manner, and etc can influence the product distribution significantly. The resemble CH_4 selectivity of the Ru based catalysts can be related to their high metal dispersion (2–5 nm); the increasing C_{5+} selectivity of the Ru-in/TNT can be associated with the space confinement effect of TNT as well.

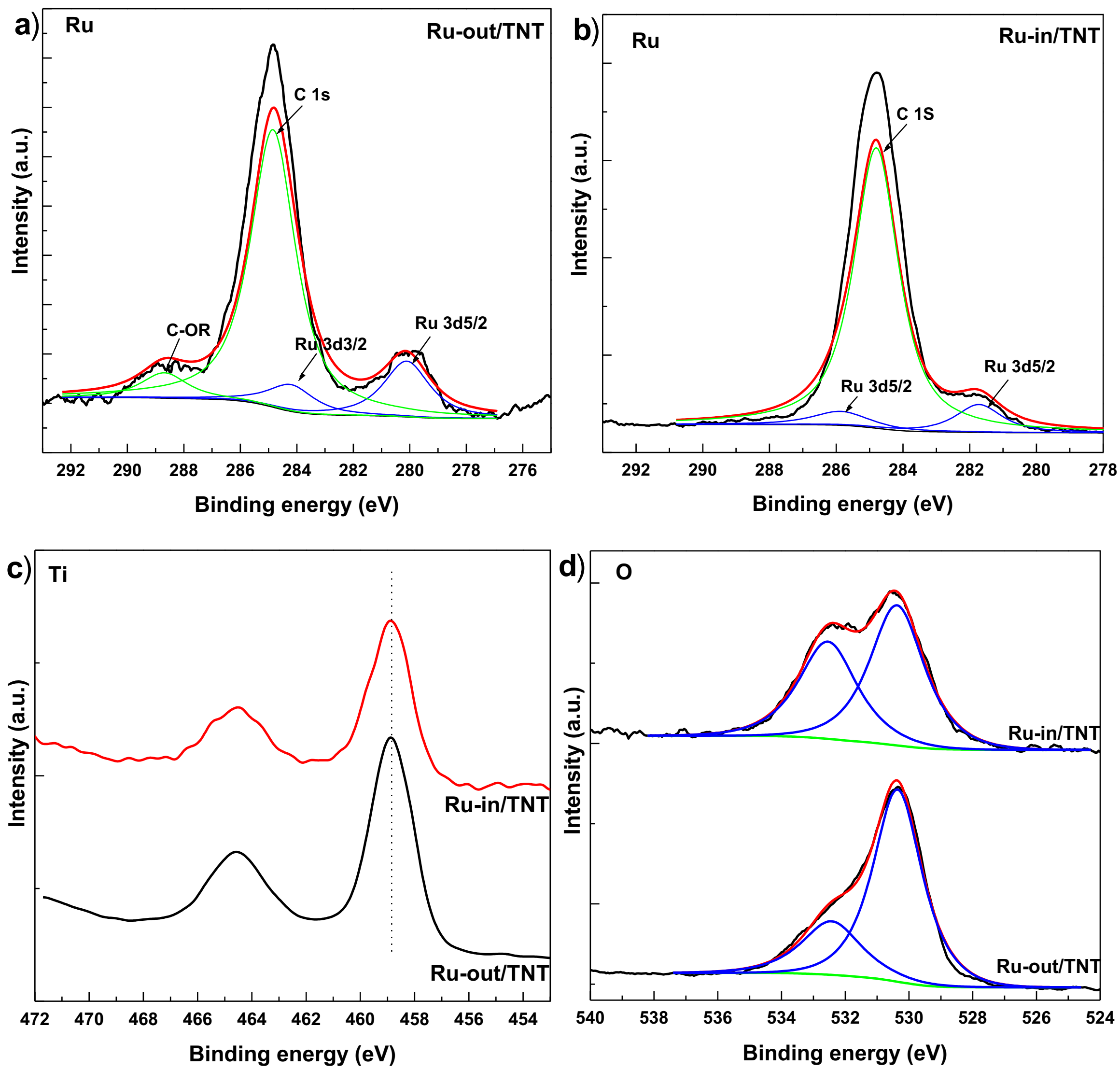

Fig. 5. XPS spectra of spend catalysts.

Table 4
Catalytic performance of different catalysts after 48 h of FTS.

Catalyst	CO Conv. (%)[a]	Activ. (μmol/g_{Ru} s)[b]	[c]Selectivity (%)				[d]O/P
			CH_4	C_{2-4}	C_{5-18}	C_{19+}	
Ru-out/TNT	21	9	22.2	19.3	57.5	1.0	0.80
Ru-in/TNT	44	60	20.3	15.0	62.5	2.2	0.65
Ru/P25	22	8	21.0	18.7	55.1	1.2	0.79

[a] CO conversion rate at the initial reaction period.
[b] Activity rate at the relatively stabilized period (>20 h).
[c] Selectivity determined at similar CO conversion of ∼20% for each catalyst.
[d] Molar ratio of olefin to paraffin.

4. Discussion

It can be seen that the highly dispersed Ru NPs can be deposited in the inner pores of TNT via vacuum-assisted impregnation route. The confinement effect of TNT on the entrapped Ru NPs is significantly, which can be discussed from the following respects

Firstly, the confinement effect of TNT can enhance the metal-support interactions, which effectively hinders the metal sintering and metal leaching through the FTS. Metal sintering and metal leaching are inevitably avoided in most part of the heterogeneous catalysis process, which is strongly affected by the support properties, loading species state, presence of other ions, and etc. [27,32]. The former can cause the growth of particles size and leads to a lower active surface area; the latter one, arising from oxidation of active metallic Ru [33], or formation of volatile species [34], intend to lead to loss of active species. Both the two processes are responsible for the catalysts' activity decline in this work. In documents elsewhere [35–38], modulating the metal-support interaction to control the metal sintering/leaching has proven to be an effective protocol. The Ru-in/TNT showed better resist-metal sintering/leaching property than did the Ru-out/TNT, as shown in Table 3 and Fig. 3. It demonstrated well that the metal sinter-

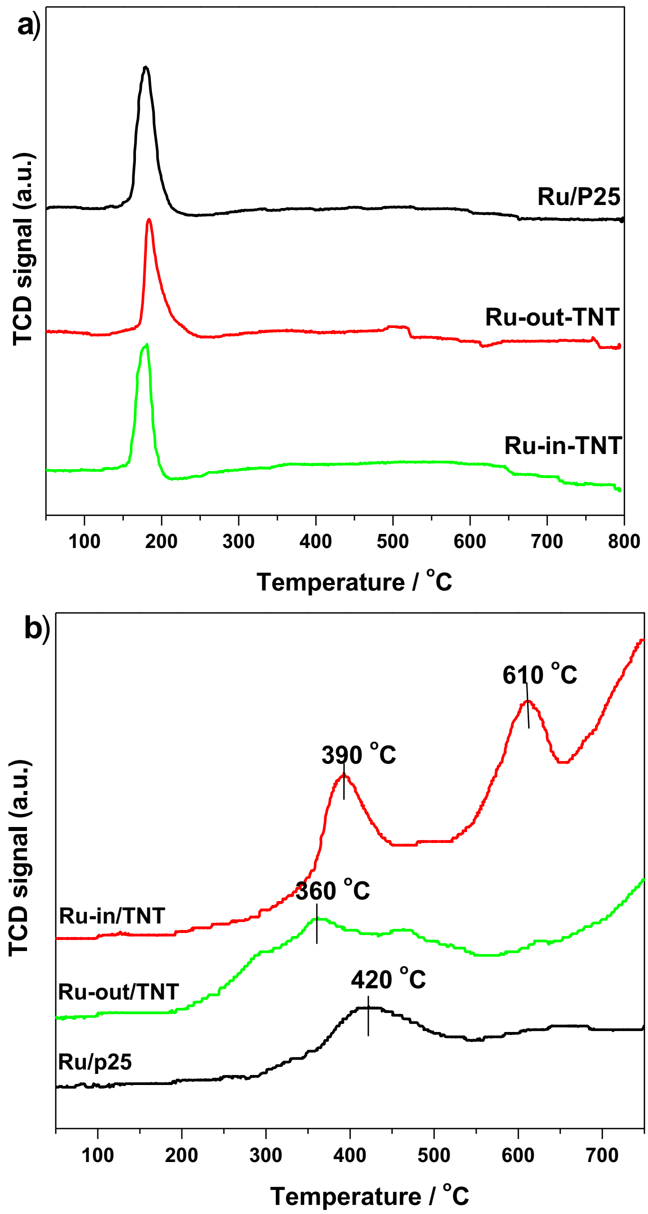

Fig. 6. H_2-TPR and H_2-TPD pattern of various catalysts.

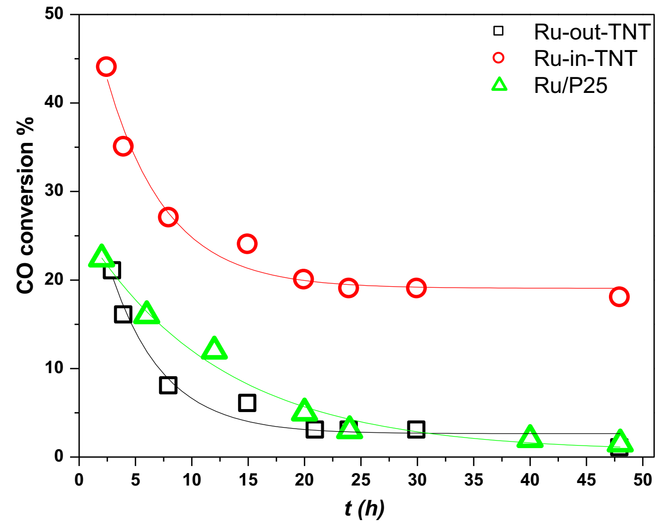

Fig. 7. CO conversion as function of time evolved for various Ru catalysts.

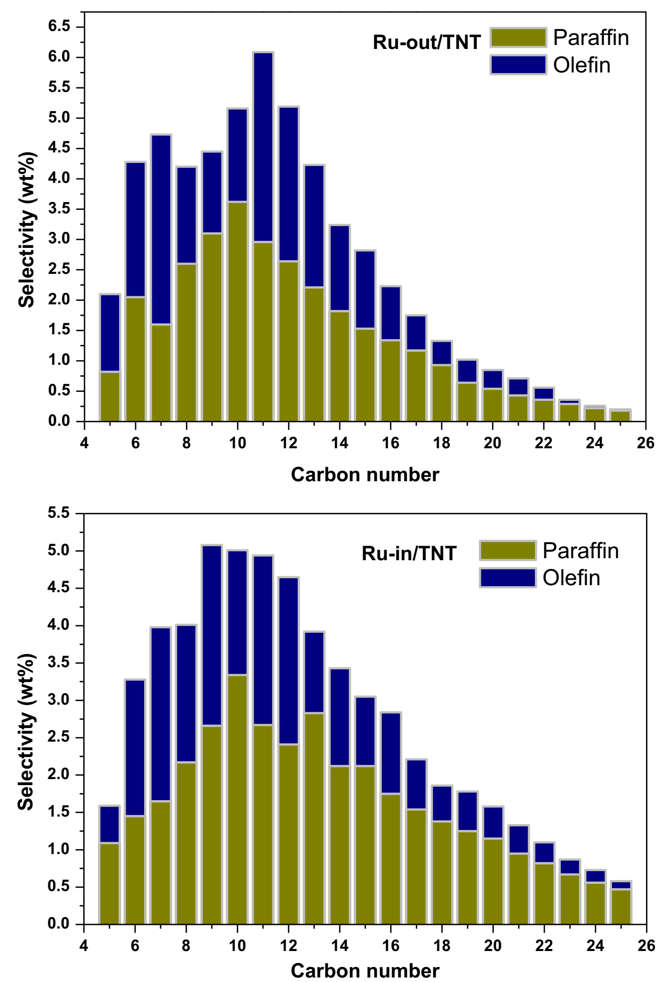

Fig. 8. Product distribution of different catalysts.

ing/leaching can be limited by the confinement effect of TNT. This can be ascribed to the intensified metal-support interaction.

Secondly, the confinement effect of TNT could promote the charge transfer between the Ru component and support, and modulate the electronic state and reducibility of the Ru components (justified by the XPS and H_2-TPR analysis). This could produce more active center sites, resulting in increased hydrogen adsorption capacity, as shown by H_2-TPD. Shetty's group [39] has investigated the mechanism of CO dissociation on ruthenium surface. They declaimed that the hydrogen assisted CO dissociation pathway is the initial step in FTS. It can be assumed that the increased hydrogen adsorption active sites could facilitate the CO dissociation on Ru NPs, leading to enhanced CO conversion activity. In Bao's group [40], they also found that the confinement effect of carbon could modulate the redox properties of entrapped FeO, promote the production of Fe_xC_y species, resulting in improved FTS activity.

Thirdly, the confinement effect of TNT can enhance the re-adsorption and re-insertion of α-olefin intermediates, which facilitates the growth of carbon-chain. Herein, we intend to ascribe the increasing selectivity towards long-chain product to the confined space of the tubular cavity. The approximate one-dimension reaction space of nanotubes can slow down the diffusion of reactants and products and arise the enrichment of intermediates, mainly α-olefin ($R{-}CH{=}CH_2$) [41]. Therefore, the probability of re-adsorption of α-olefin intermediates [42] can be enhanced. And by combining with methylene (-CH_2−) or other α-olefin, short-chain products can grow continually, resulting in the increase of C_{5+} products. This space confined phenomenon is also found in former work about Fe entrapped inside CNTs [11] and TNTs [43].

5. Conclusion

A high performance Ru-in/TNT catalyst with Ru NPs entrapped in TNT was prepared by valcumm-assisted impregnation method. During FTS synthesis, the Ru-in/TNT presents enhanced CO conversion, deactivation-resist ability, as well as increasing selectivity towards long-chain product. The high performance can be ascribed to the confinement effect of TNT on the entrapped Ru NPs: strengthen metal support interaction, promoted charge transfer and enhanced re-adsorption of α-olefin intermediates. The encouraging results suggest that the TNT confined Ru NPs catalyst can be promising catalyst to improve the FTS performance.

Acknowledgements

This work was supported by the National Scientific Foundation of China (project No. 21303210), and the National 973 Project of China (project No. 2013CB228105).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.apcata.2016.07.021.

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Supporting information

Effect of confinement of TiO_2 nanotubes over the Ru nanoparticles on Fischer-Tropsch synthesis

Xu Yang[1], Wenbo Wang[1], Liangpeng Wu[1], Xinjun Li[1], Tiejun Wang[1*] Shijun Liao[2]*

[1]Key Laboratory of Renewable Energy, Guangzhou Institute of Energy Conversion, Chinese Academy of Sciences, Guangzhou 510640, China

[2]School of Chemistry and Chemical Engineering, South China University of Technology, Guangzhou 510641, China

[1] Corresponding authors: Prof. Li Xinjun, lixj@ms.giec.ac.cn;
Prof. Wang Tiejun, wangtj@ms.giec.ac.cn

Table S1 FTS performance of Ru-in-TNT reacted in various temperatures.

Temp. (°C)	[a] Conver. (%)	[b] Select. (%) CH_4	C_{2-4}	C_{5+}
200	25	20.6	15.0	64.4
220	32	22.7	14.4	62.9
240	44	22.4	15.6	62.0
260	61	24.5	12.0	63.5

[a] CO conversion rate at the initial reaction period

[b] Selectivity determined at similar CO conversion of 20% for each run

Table S2 The specifications of fresh catalysts.

Catal.	[a]Mean particle size (nm)	[b]Metal dispersion	[b]Surface Ru sites (mmol g^{-1})	[c] TOFs (10^{-3} s^{-1})
Ru-out/TNT	3.2	0.41	0.204	20
Ru-in/TNT	2.8	0.47	0.224	36

[a] average particle size calculated from the TEM image by recording randomly selected 200 particles .

[b] the metal dispersion and surface Ru sites were determined using equation (1) and (2), respectively[1].

$$D=1.32/d \qquad (1)$$

$$[Ru]_{surf}=[Ru]^0\times D \qquad (2)$$

Where the D, d, $[Ru]_{surf}$ and $[Ru]^0$ stand for the metal dispersion, mean particle diameter, surface Ru sites and total amount of Ru, respectively.

[c] Evaluated from the rate of CO conversion at the initial 2h reaction period per Ru atom on the surface

Table S3 The specifications of spend catalysts.

Catal.	Mean particle size (nm)	Metal dispersion	Surface Ru sites (mmol g^{-1})	[a] TOFs (10^{-3} s^{-1})
Ru-out/TNT	5.5	0.24	0.028	17
Ru-in/TNT	4.0	0.33	0.134	31

[a] Evaluated from the rate of CO conversion at 24 h reaction per Ru atom on the surface

Table S4 The amount of Ru in the spend catalysts and liquid phase of product.

Catalyst	[a] Ru concentration (mmol g^{-1})		
	C_{bulk}	C_{liquid}	C_{Total}
Ru-out/TNT	0.12	0.35	0.47
Ru-in/TNT	0.41	0.08	0.49

[a] The C_{bulk} and C_{liquid} were referred as the Ru concentration in the solid bulk of used catalyst, in the liquid phase of FTS product, respectively, which were determined by the ICP technique. The C_{Total} was defined as the sum of C_{bulk} and C_{liquid}.